

Trends of carbon emission intensity from electricity energy generation across Italian regions

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Data Management Project Presentation 2025-26

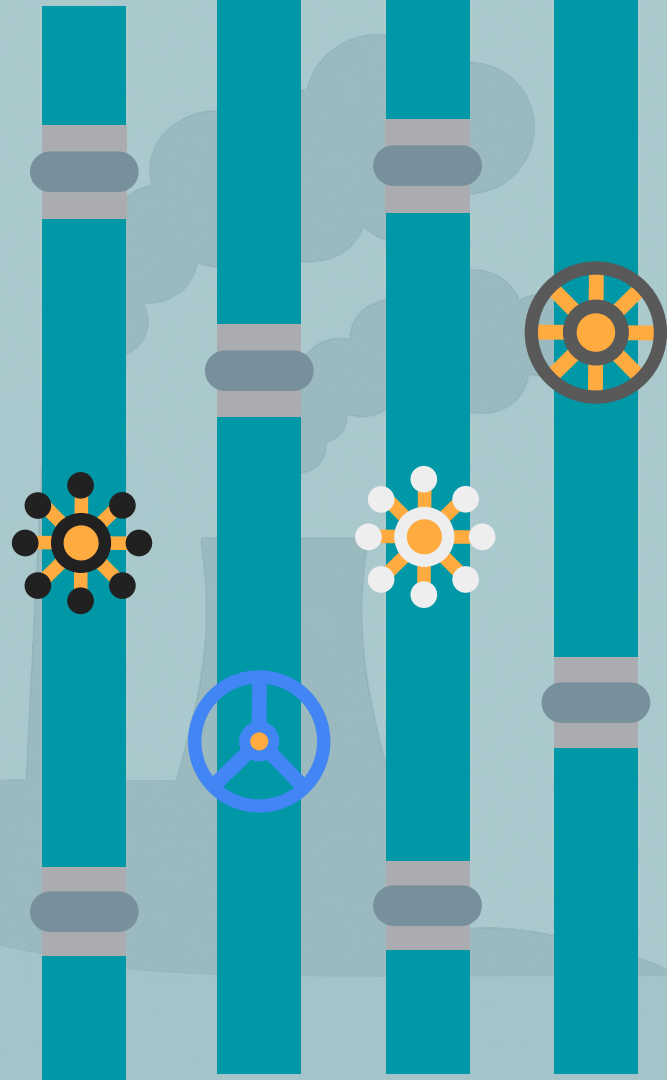
RESEARCH QUESTION

The aim of the project is to **compare greenhouse gas emissions between Italian regions**. In particular, we developed the **carbon emission intensity (CEI)** indicator, a very simple but valid metric for the comparison.

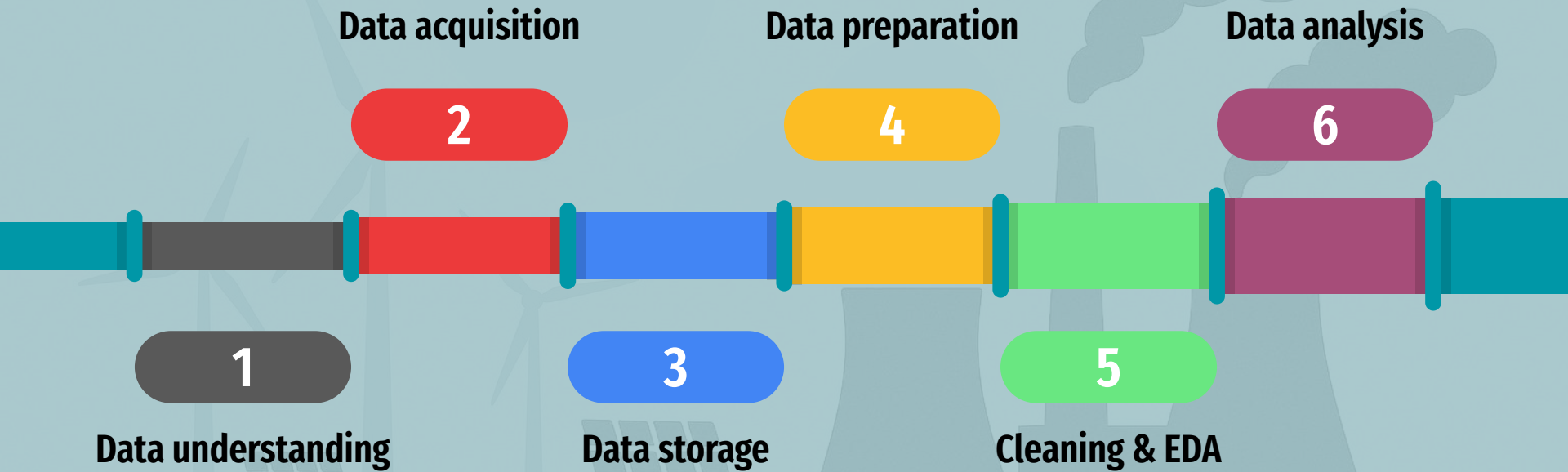
$$C_{y,r} = \frac{\text{Emissions from energy generation}_{y,r}}{\text{Energy generation}_{y,r}}$$

We choose to use **one specific sector (emissions due to electricity generation)** because regions vary greatly in many terms (population, transport, climate, ...) while power plants are needed everywhere and the electricity generated can be used as a factor of normalization for the comparison.

The indicator becomes a proxy for the use of renewable sources and the **effectiveness of decarbonization policies**.



DATA PIPELINE



1 DATA UNDERSTANDING

1

Terna

Italy's transmission system operator (TSO)

Yearly Italian regional electricity generation by source (GWh)



2

EDGAR

European Emissions Database for Global Atmospheric Research

Yearly European regional emissions by sector (ktco2eq)



3

ISPRA

Italy's Istituto Superiore per la Protezione e la Ricerca Ambientale

Italian regional emissions from energy sector for certain years (tco2eq)



2 DATA ACQUISITION

1

Terna

from Terna API (JSON)

- authorization /sleep function
- merged many JSON in single one
- remapped to table with schema of Terna documentation

13704 records x 6 columns

year: from 2005 to 2022

production type: “Netta” / “Lorda”

region: 20 Italian regions

province: Italian provinces

source: “Fotovoltaico”, “Idrico”, “Termoelettrico”, “Eolico” and “Geotermoelettrico”

production (GWh): amount of electricity production

2 DATA ACQUISITION

2

EDGAR

from download (.xlsx)

-removed non-Italian data

125 records x 37 columns

substance: GWP_100_AR5_GHG

ISO: e.g. ITA **Country:** e.g. Italy, San Marino and the Holy See

NUTS2: e.g. ITC3 **NUTS2 desc:** e.g. Liguria

sector: Dom_Avi, Dom_Ship, Indirect, Agriculture, Buildings, Energy, Industry, Transport and Waste

Y_XXXX: emissions (in ktco2eq) of year XXXX from 1990 to 2022

2 DATA ACQUISITION

3

ISPRA

from download (.xlsx)

-took Valle d'Aosta for enrichment

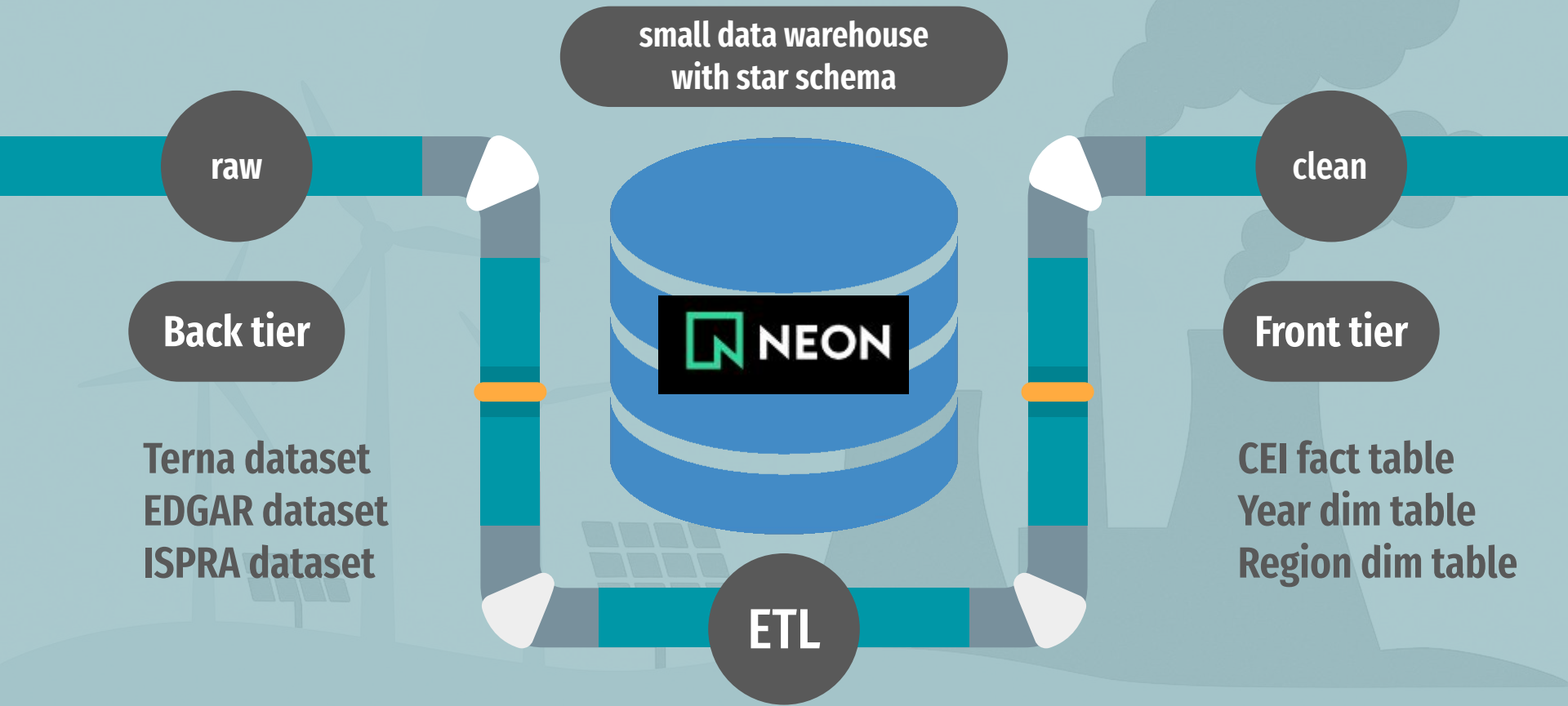
40 records x 10 columns

Regioni: 20 Italian regions

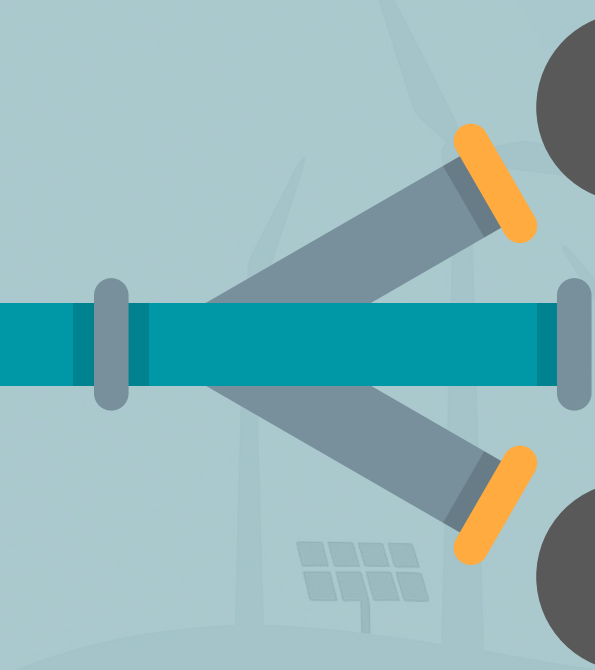
XXXX: energy sector emissions of the following XXXX years
(in tco2eq):

1990 1995 2000 2005 2010 2015 2017 2019 2021

3 DATA STORAGE



4 DATA PREPARATION: INTEGRATION



Schemas transformation

Choosing schema structure

Wanted a final table with year, region and CEI indicator.
Pivoting EDGAR years to obtain same schema.

Correspondences investigation

Conflicts between datasets

Name, type and model heterogeneities (e.g. different units of measure, homonyms and synonyms, misclassification of Italian regions across datasets...).

Schemas integration

Mapping rules

Renaming attributes and keys, pivoting tables, deciding over null values. Obtained a dataset with 20 regions x 18 years = 360 measures.

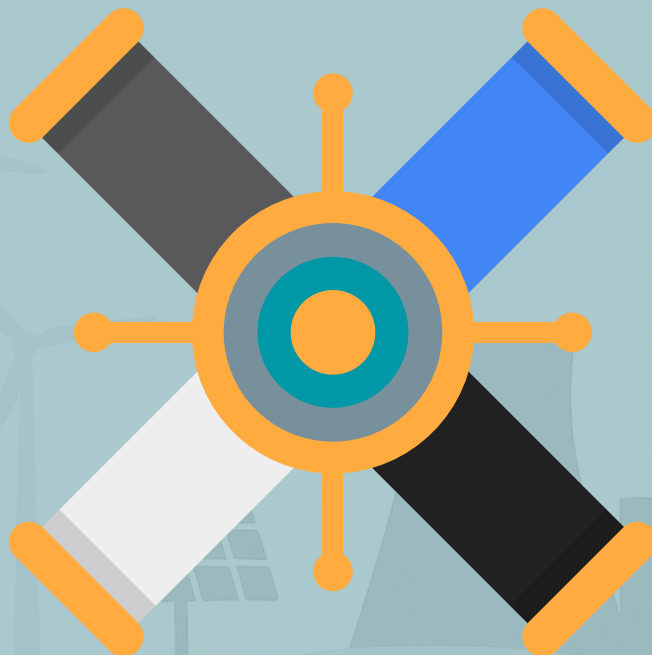
4 DATA PREPARATION: INTEGRATION

Time dimension

Selecting 2005-2022 years
to cover a common time span
before and after the start of
decarbonization policies

Space dimension

Aggregating provinces
since only interested in
regional level and wanted
compatible space granularity
across datasets



Source dimension

Aggregating all sources

since only interested in total
electricity generation

Sector dimension

Selecting “Energy” sector
since only interested in
emissions from electricity
production

4 DATA PREPARATION: DATA QUALITY

Semantic accuracy

Searching for the meaning of dimensional attributes (e.g. regions and years) during integration. It was important to understand EDGAR sectors meaning.



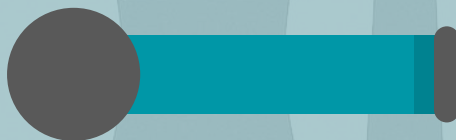
Syntactic accuracy

Checking for differences in the nomenclature between datasets (e.g. Valle d'Aosta vs Valle d'Aosta/Vallée d'Aoste).



Completeness

Evaluating the number of missing values for each table and assessing the representativeness of the data with respect to the phenomenon and dimensions considered.



Other quality dimensions

In the report we referenced to the **coherence** of data for our purpose, the **redundancy** of certain columns and **correctness** of data due to the trustable sources we used.



5 DATA CLEANING & EDA

Dataset statistics	
Number of variables	6
Number of observations	13704
Missing cells	0
Missing cells (%)	0.0%
Duplicate rows	0
Duplicate rows (%)	0.0%

Terna

Dataset statistics	
Number of variables	3
Number of observations	360
Missing cells	0
Missing cells (%)	0.0%
Duplicate rows	0
Duplicate rows (%)	0.0%

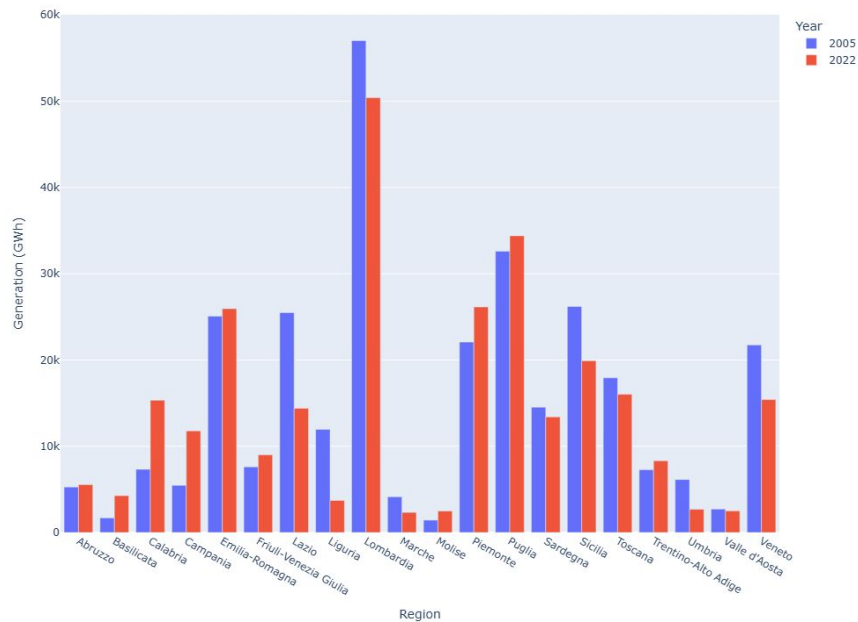
Dataset statistics	
Number of variables	37
Number of observations	125
Missing cells	0
Missing cells (%)	0.0%
Duplicate rows	0
Duplicate rows (%)	0.0%

EDGAR
enriched
with ISPRA

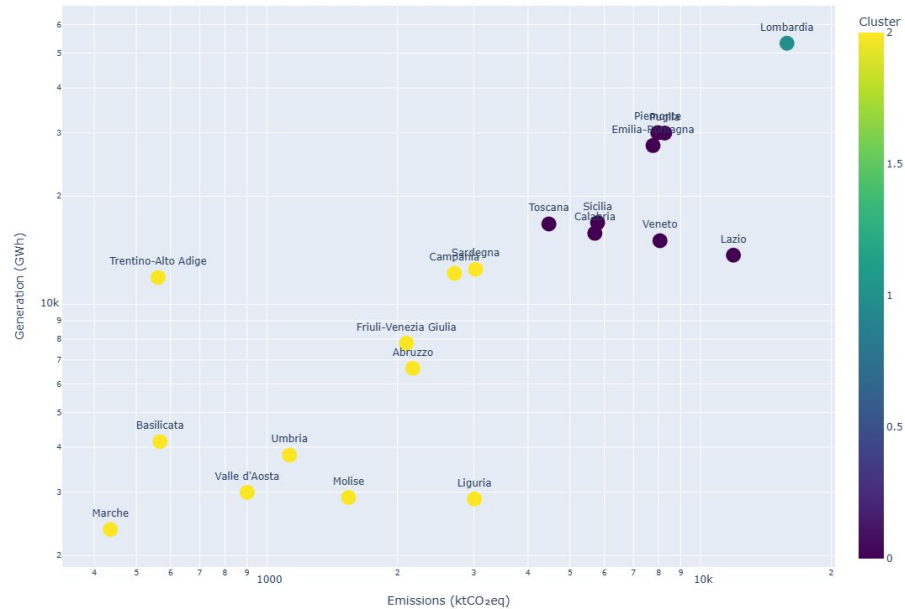
Dataset statistics	
Number of variables	3
Number of observations	360
Missing cells	12
Missing cells (%)	1.1%
Duplicate rows	0
Duplicate rows (%)	0.0%

5 DATA CLEANING & EDA

Side-by-side comparison of regional net generation: 2005 vs 2022



Clustering of Regions - 2021 Generation vs Emissions - Log scale



5 DATA CLEANING & EDA

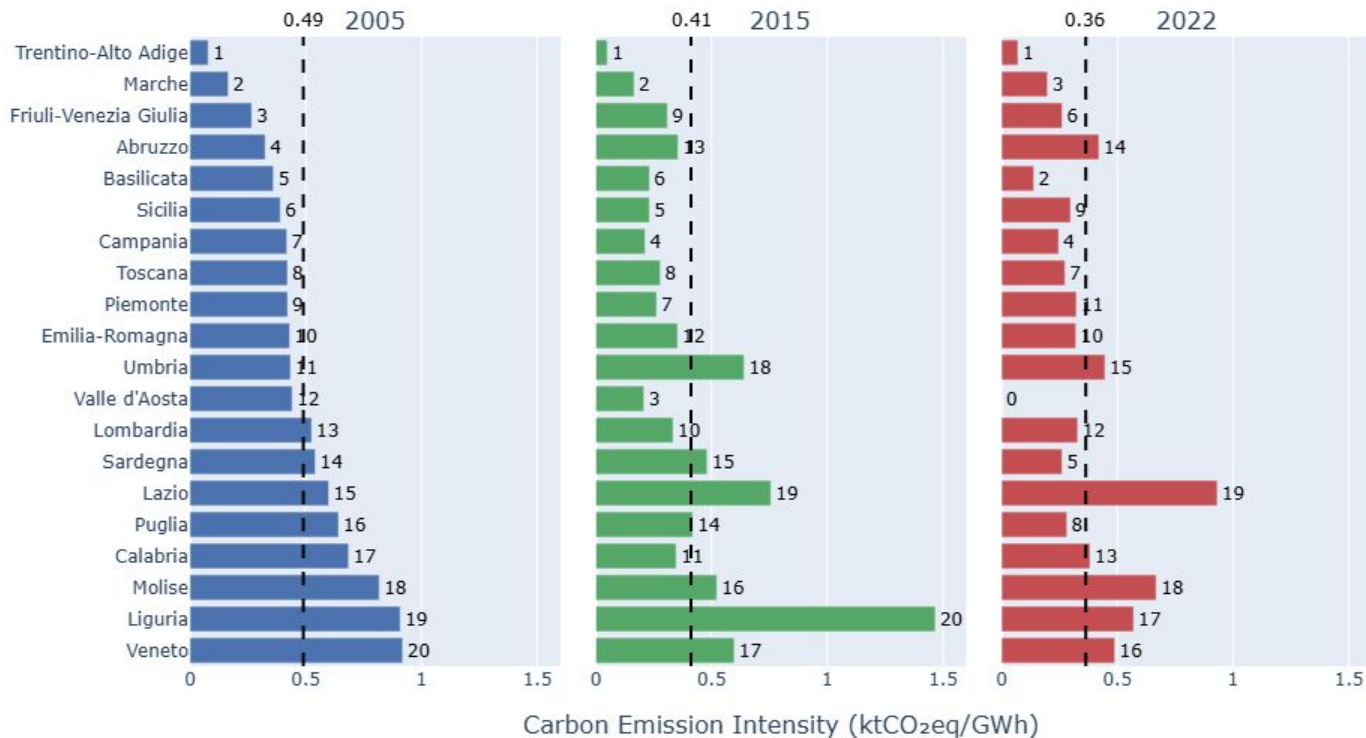
☐ id integer	☐ Year integer
☐ 0	2005
☐ 1	2006
☐ 2	2007
☐ 3	2008
☐ 4	2009
☐ 5	2010
☐ 6	2011
☐ 7	2012

☐ id integer	☐ Region text
☐ 0	Abruzzo
☐ 1	Basilicata
☐ 2	Calabria
☐ 3	Campania
☐ 4	Emilia-Romagna
☐ 5	Friuli-Venezia Giulia
☐ 6	Lazio
☐ 7	Liguria

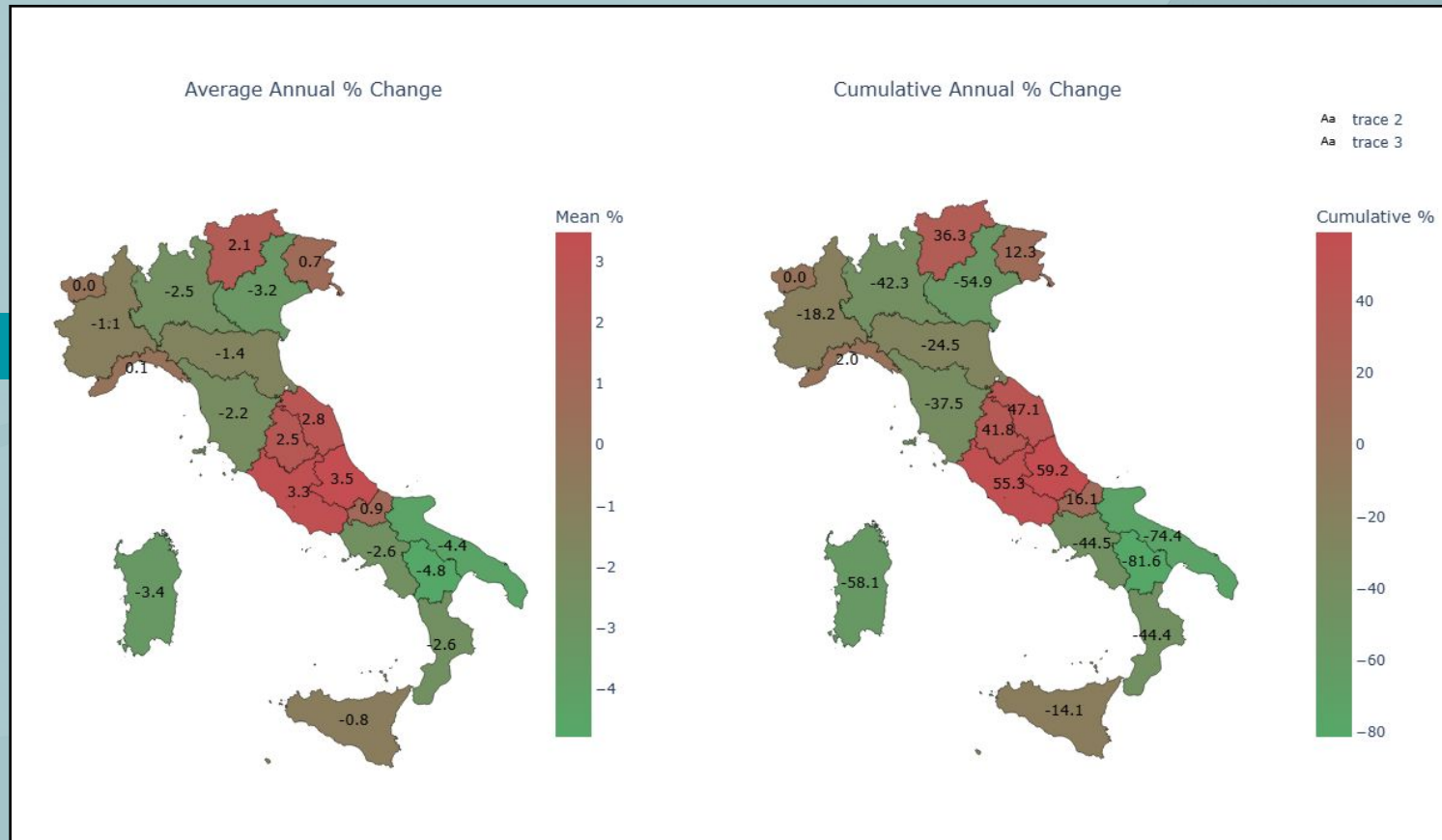
☐ year_id integer	☐ region_id integer	☐ carbon_emission_inten...
☐ 0	0	0.3250390241662307
☐ 0	1	0.36049258848630816
☐ 0	2	0.68534191817133
☐ 0	3	0.4185158001766615
☐ 0	4	0.43096745395317715
☐ 0	5	0.2666909259421016
☐ 0	6	0.599440739508853
☐ 0	7	0.9083239994781448
☐ 0	8	0.5255310120725201

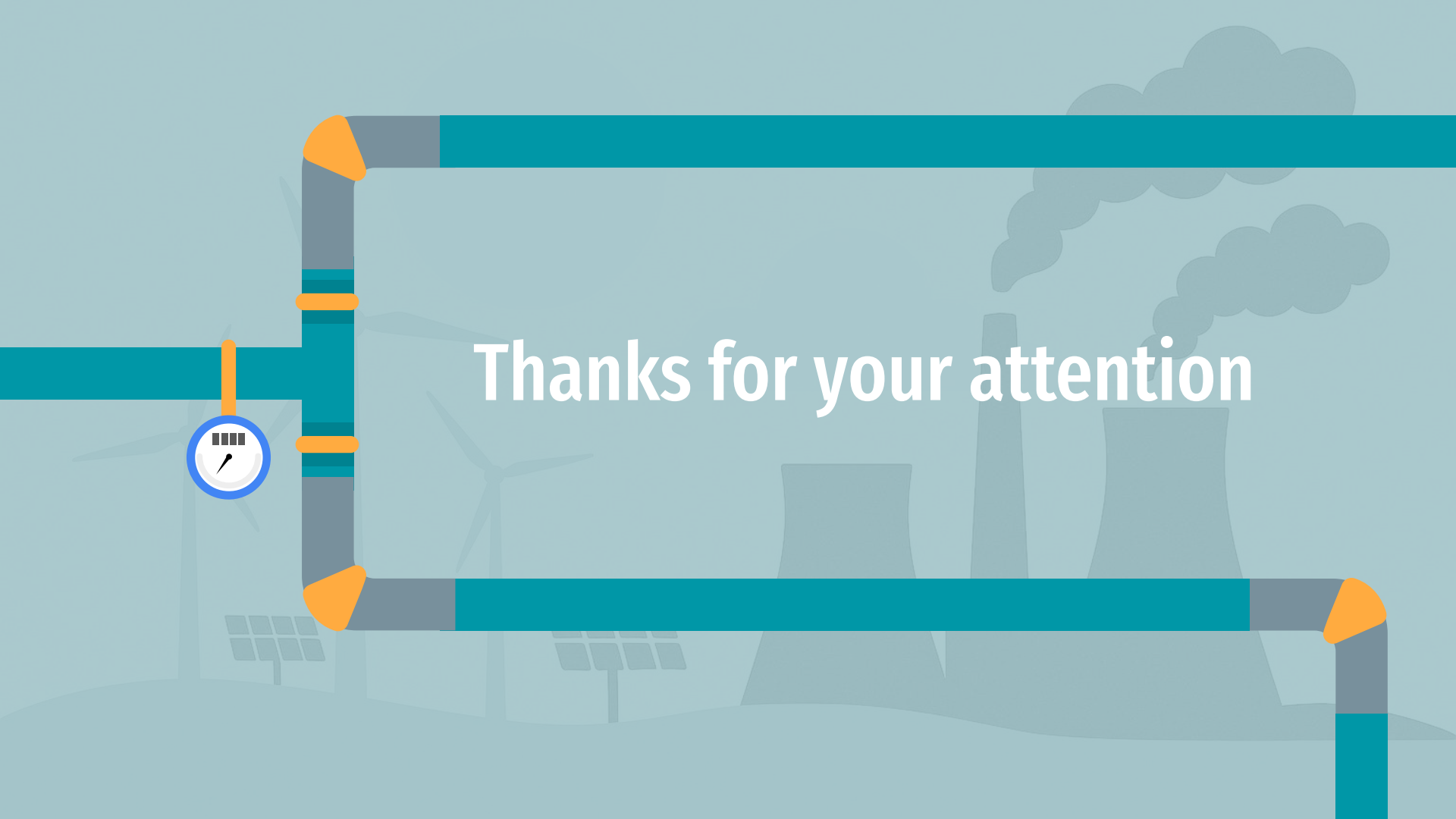
6 DATA ANALYSIS

Regional emissions ranking: 2005 vs 2015 vs 2022



6 DATA ANALYSIS





Thanks for your attention